Where and How Does a Frozen Pixel-Based World Model Represent Calibrated Posterior Uncertainty About Occluded Objects?

Relevant Past Papers

- **Zhang, "What Do World Models Learn in RL? Probing Latent Representations in Learned Environment Simulators" (arXiv:2603.21546)**. Linear and MLP probes plus causal interventions on IRIS and DIAMOND (Atari) show that game-state variables are approximately linearly decodable and functionally used by generation. Its limitation is that it decodes point estimates of visible objects only with no posterior uncertainty, no Bayesian-oracle calibration, and interventions that are generic rather than belief specific. This is the closest analog and our primary differentiation target.
- **Li, Hopkins, Bau, Viégas, Pfister, Wattenberg, "Emergent World Representations" (Othello-GPT, ICLR 2023, arXiv:2210.13382)**. A GPT trained only on legal-move prediction develops an emergent internal board-state representation that interventions can use to control its output; its limitation is that the domain is discrete and fully determined, so there is no partial observability, no uncertainty, and therefore no calibrated belief to speak of.
- **Nanda et al., "Emergent Linear Representations in World Models of Self-Supervised Sequence Models" (arXiv:2309.00941)**. The Othello board turns out to be linearly decodable in a "MINE vs. YOURS" basis and steerable by simple vector arithmetic, establishing linear state representations as a real and manipulable phenomenon; its limitation is that the target remains a point estimate state with no uncertainty semantics.
- **Shai, Marzen, Teixeira, Oldenziel, Riechers, "Transformers Represent Belief State Geometry in their Residual Stream" (NeurIPS 2024, arXiv:2405.15943)**. Belief states are linearly represented in transformer residual streams even when the predicted belief geometry is highly nontrivial and fractal, derived via the mixed-state presentation of the data-generating process; its limitation is that the evidence comes from abstract synthetic HMM token streams (e.g., Mess3) rather than a pixel-based action-conditioned world model, with no occluded physical object, no calibration against a tracking posterior, and no generative steering. This paper supplies our core conceptual justification.
- **Chen, Viégas, Wattenberg, "Beyond Surface Statistics: Scene Representations in a Latent Diffusion Model" (arXiv:2306.05720)**. Linear probes recover depth and salient-object/background maps early in the denoising process, and interventions on those representations causally affect the generated image; its limitation is the static-scene setting, with no temporal belief updating, no uncertainty, and no occlusion dynamics.
- **Esmati, Nath, Hofmann, Nowrouzezahrai, Ebrahimi Kahou, Mirmehdi, "The Invisible Hand of Physics: When Video Diffusion Models Know More Than They Show" (arXiv:2606.05328)**. Inverting the sampling ODE reveals that physical plausibility is

linearly decodable from video diffusion internals and outperforms V-JEPA and VideoMAE. Its limitation is that this is a binary plausibility read-out rather than a continuous posterior over an occluded object's position, with no calibration, no oracle, and no steering.

- **Garrido, Ballas, Assran, Bardes, Najman, Rabbat, Dupoux, LeCun, "Intuitive physics understanding emerges from self-supervised pretraining on natural videos" (arXiv:2502.11831)**. V-JEPA reliably separates physically plausible from implausible videos under the violation-of-expectation paradigm while pixel and text-based models sit near chance; its limitation is that this is a behavioral surprise metric rather than a decoded internal belief, and it says nothing about calibrated uncertainty over a specific hidden location.
- **"Causal Physics Steering in Video World Models via Concept Activation Vectors" (arXiv:2605.24322)**. Training-free signed CAV injection flips object-permanence plausibility judgments across an α-sweep, demonstrating that physics concepts exist as causally manipulable directions; its limitation is that it steers a binary plausibility score rather than posterior location, and it operates on an encoder (VideoMAE) that does not emit steered frames.
- **Newell, Hu, Lipson, Richter, Koltun, "CoMotion: Concurrent Multi-person 3D Motion" (ICLR 2025, arXiv:2504.12186)**. A recurrent model updates 3D poses directly from pixels, tracking through occlusion and snapping back to the correct state on reappearance, proving that occlusion tracking is learnable from pixels; its limitation is that it is a supervised tracker with an explicit pose state and no calibrated positional uncertainty, and its hidden state is never probed.
- **Upadhyay et al., "WorldBench: Disambiguating Physics for Diagnostic Evaluation of World Models" (arXiv:2601.21282)**. A concept disentangled video benchmark scoring world models on fine-grained physical concepts including object permanence and occlusion; its limitation is that it evaluates behavioral output only, with no internal representation probing and no calibrated belief target, which is precisely the gap MineOcclude fills.

Motivation

- Interactive world models such as MineWorld, Oasis, and Genie are increasingly proposed as "simulators" for planning and agent training. But a simulator that must render the world after an object is hidden faces an inherently partially observed problem: the correct behavior is not to remember a single location, but to maintain a distribution over where the object could be, widen it appropriately as time passes and the object could have moved, and collapse it when the object reappears. Whether today's frozen pixel world models actually do this, as opposed to storing a brittle point estimate that decays into forgetting or hallucination, is unknown. Existing probing work establishes that world models encode point-estimate state of visible objects (arxiv 2603.21546 and 2309.00941) and that video models can classify physical plausibility (arxiv 2606.05328), but no one has asked whether a pretrained pixel world model holds a calibrated Bayesian belief about a hidden object.

- Multiple 2025–2026 papers document that autoregressive world models "forget or hallucinate objects that were visible in earlier frames, particularly severe in long-horizon rollouts" (arXiv:2606.18960) and motivate memory modules precisely because models forget hidden objects once they leave the context window (arXiv:2512.06983). This tells us the capacity to maintain hidden-object information is fragile, but not whether the residual stream encodes a probability distribution with the right spread. That is an internal-representation question that only probing + causal intervention can answer.
- **Importance**
  - **(1) Safety/planning:** an agent that plans in a world model needs the model's uncertainty about occluded hazards to be calibrated, or it will be overconfident exactly where it is blind.
  - **(2) Science of world models:** distinguishing point-estimate state from calibrated posterior uncertainty is the sharp version of the "do models have world models?" question, and connects the empirical probing literature (Othello GPT, IRIS/DIAMOND) to belief-state theory.
  - **(3) Interpretability:** a calibrated, causally-active uncertainty direction is a concrete, falsifiable target.
- We (i) construct scenes where the exact Bayesian posterior over the hidden object is analytically known (a scripted random walk with a closed-form Kalman posterior), (ii) decode the world model's internal posterior mean and log-uncertainty with cross-validated probes, (iii) grade the decoded uncertainty against the oracle with a calibration battery, and (iv) causally steer generation along an uncertainty-targeted direction. Transformers trained on next-token prediction linearly represent Bayesian belief geometry in the residual stream. And we know world model states are linearly decodable and causally used. Our contribution is to raise the bar from state to calibrated posterior uncertainty.

Key Ideas/Contributions/Novelty

Research questions: For a frozen, pixel-based autoregressive world model (MineWorld), does it internally represent calibrated posterior uncertainty about an occluded object's state? Where in the network and how (geometrically) is that representation computed? Why does it succeed or fail as occlusion duration approaches and exceeds the architectural context limit?

Representational ladder: For latent object state s_t under partial observability, we distinguish:

- **Level 0 State (s_t):** the ground-truth hidden state (position/presence), available only to our oracle.
- **Level 1 Point-estimate memory/extrapolation ($\hat{x}_t$):** a single decodable best guess of the hidden object's location (posterior mean only).
- **Level 2 Calibrated posterior uncertainty representation (the target):** an internal representation whose decoded spread tracks the uncertainty an ideal Bayesian observer should have. It is essentially like the POMDP filtering posterior with a decodable posterior

mean, a calibrated posterior spread, correct update dynamics (should grow as time under occlusion increases), and causal use of the estimate.

Criteria for Uncertainty:

- **U1 (Decodable posterior mean):** A probe recovers the hidden object's location above chance and above a last-seen baseline ($R^2 > 0$ vs. shuffled controls at $p < 0.01$, and > last-seen baseline).
- **U2 (Calibrated posterior uncertainty):** The decoded spread tracks the oracle posterior spread conditional on elapsed time: partial correlation $\rho(\hat{\sigma}, \sigma \mid \tau) > 0$ with a bootstrap CI excluding 0; plus a full calibration battery (reliability/ECE on interval coverage, and z-score coverage). This is the calibration test in the sense of (arxiv 1706.04599), applied to a decoded posterior rather than a softmax head.
- **U3 (Correct uncertainty dynamics):** Uncertainty grows at approximately the oracle rate during occlusion, snaps back on reveal, and reflects absence as evidence (uncertainty stops growing / re-centers when the object is provably not where expected).
- **U4 (Causal/functional use):** A signed dose-response of reappearance location under injection of the uncertainty/mean direction with specificity controls and SSIM coherence. This would demonstrate that the represented estimate is used and not just present.

Contributions:

1. **A mechanistic explanation of where, how, and why the model represents (or fails to represent) calibrated posterior uncertainty about occluded objects.** We localize the computation across depth, characterize its geometry (angles among the decoded position, uncertainty, and elapsed-time directions), tie decoding collapse causally to the context boundary, localize which tokens carry the hidden object information, and diagnose the cause of each failure mode (e.g., "frozen last-seen snapshot" vs. "mean-only tracking" vs. "clock substitution"). This is the paper's primary product
2. **A representational ladder with falsifiable uncertainty criteria U1–U4 and a POMDP oracle grading protocol** that make the "calibrated posterior uncertainty" claim measurable and the mechanistic verdicts are capable of being settled.
3. **An occlusion benchmark of controlled trajectories** with matched Bayesian-filter oracle posteriors, enabling reproducible study of uncertainty representation under partial observability.

Methods

- MineWorld is a visual-action autoregressive transformer with a LLaMA-style decoder over interleaved discrete image tokens (VQ image tokenizer, fine-tuned from a pretrained checkpoint on VPT data) and action tokens. We will use the 700M_16f checkpoint as the core model and keep 300M_16f and 700M_32f for checkls.
- **Pipeline:**

- Minecraft server (superflat, fixed light/weather, no spawn) → /summon magenta sheep (NoAI, Silent, PersistenceRequired, Color:2)
- Scene scripting:
  - Regime A: RCON teleport @20Hz, drifting random walk (known v,q)
  - Regime B: AI-on sheep in fenced lane (dynamics fit from logs)
- record frames + actions + JSONL {tick, sheep_x/z, vel, player pose, rng_seed, visible_flag(geometric ray-wall $\wedge$ empirical magenta)}
- Bayesian ORACLE:
  - Regime A: closed-form Kalman posterior $\mu(\tau)$, $\sigma^2(\tau)$
  - Regime B: particle filter (M=10k, absence-as-evidence, systematic resample ESS<M/2); sensitivity ×2, ×0.5 noise
- FROZEN MineWorld (teacher-forced over recorded frames+actions) → forward hooks → per-layer mean-pooled visual-token activations (+ token-level at 2–3 layers)
- PROBES:
  - ridge regression (z-scored, 5-fold CV grouped by scene layout) → decode $\hat{\mu}$ (U1) and log $\hat{\sigma}$ (U2)
  - 2-layer MLP (256→128→1); $\Delta = R^2(\text{MLP}) - R^2(\text{linear})$ = linearity/measurement test
- CALIBRATION battery (U2,U3) + CAUSAL intervention (U4)

- **Scene construction:** Superflat world, fixed lighting/weather, spawning off, ~40×20-block arena, one stone-wall occluder 3 blocks tall of variable length (controls occlusion duration), one high-contrast object: a single magenta-dyed sheep (/summon ... {NoAI:1,Silent:1,PersistenceRequired:1,Color:2}). Camera fixed ~10–14 blocks back with small natural movements, returning to a canonical pose ~1 s before reveal to preserve homography validity. Sheep moves mostly along one axis parallel to the wall.
- **Motion Regimes:**
  - **Regime A (core, exact oracle).** NoAI sheep moved by script at 20 Hz via RCON teleportation with a drifting random walk with known drift v and process noise q. The occlusion posterior is closed-form. Because constant velocity extrapolation equals the oracle mean here, it is a reference, not a competitor; the meaningful ceiling is an explicit feature regression given [last-seen position, estimated speed, elapsed time].
  - **Regime B (robustness).** AI motion on sheep wandering in a fenced lane; dynamics fit from 30–60 min calibration logs. Oracle is a particle filter with a sensitivity check re-running at noise ×2 and ×0.5. Constant velocity is a genuine baseline here.
- **Condition grid:** Occlusion durations $\approx\{0.1, 0.25, 0.5, 1.0, 1.5\}\times$ the 16-frame context. Speed×noise decoupling grid ($v \in \{1.0, 2.0\}$ blocks/s × $q \in \{0.02, 0.05, 0.2\}$ blocks²/s) makes true uncertainty differ at the same elapsed hidden time (powers U2). Controls: occluder only (no sheep ever) and fully visible no-wall positive control (the 3D Atari style replication).

- **Probes:** Ridge regression (z-scored, 5-fold CV grouped by scene layout, penalty chosen on held-out data) decoding oracle μ (U1) and log σ (U2); the MLP Δ serves as both linearity measure and measurement-failure diagnostic.
- **Causal intervention (U4):** Unit posterior-mean direction with the time-probe direction projected out so effects cannot be "advancing the clock." At the best layer during occlusion, add $h' = h + \alpha \cdot w_\perp$ to every visual token of the final context frame (mean pooled bookkeeping stays exact); α dosed to intended shifts of ±{0.5, 1, 2, 4} blocks plus α = 0, capped at $\|\alpha \cdot w_\perp\| \leq 0.25\times$ typical activation norm. K = 8 rollouts per (event, α) over 50 held-out events on unseen layouts (~3,600 clips). Detection: HSV magenta threshold → largest blob centroid → homography to world coordinates (4 marker blocks, OpenCV findHomography; localization error ≪ 0.5 blocks, validated on visible frames). Metrics: signed dose response slope with bootstrap CI, permutation sign tests, specificity controls (norm-matched random directions, orthogonal direction, wrong layer, wrong timestep), SSIM coherence (pushed-vs-unpushed no larger than unpushed-vs-unpushed variation). Precondition: measure the baseline persistence rate; if low, that is itself a headline result, and intervention analysis runs only on persisting clips.
- **Explanatory/mechanistic analyses:** All reuse already-extracted activations and fitted probes.
  - **Layer-wise emergence (WHERE):** probe R^2 for $\hat{\mu}$ and $\log \hat{\sigma}$ vs. depth — at what depth mean and spread become decodable and whether spread lags mean (the signature of a derived quantity); compared against (arxiv 2602.07050)'s "Physics Emergence Zone."
  - **Subspace geometry (HOW):** pairwise principal angles among $\hat{w}_\mu$, $\hat{w}_\sigma$, $\hat{w}_\tau$ per layer; $\hat{w}_\sigma \approx \hat{w}_\tau$ exposes a relabeled clock. Report angle-vs-depth curves and the $\hat{\sigma}$ residual after projecting out $\hat{w}_\tau$.
  - **Context-boundary analysis (architectural cause):** lock decoding/calibration collapse to the exact context edge (16-frame window for 700M_16f; effective context reported at 15–16 frames). Sharp degradation exactly when the last-seen frame exits context attributes miscalibration to the architectural memory limit rather than saturating Bayesian competence.
  - **Token-level localization (storage mechanism):** per-spatial-token probes at 2–3 layers. This aims to answer if hidden-object information is parked at occluder-region tokens, at last seen position tokens, or propagated frame to frame?
- **Confounds and the belief criterion they threaten:**
  - **Clock confound** ($\hat{\sigma}$ tracks only elapsed time, not true uncertainty) → threatens U2; handled by the speed×noise decoupling grid + partial correlation $\rho(\hat{\sigma}, \sigma | \tau)$.
  - **Visual leakage** (magenta partially visible, homography drift) → threatens U1/U3; handled by dual visibility labels (geometric ∧ empirical) and canonical-pose reset before reveal.
  - **Generic damage** (interventions just corrupt generation) → threatens U4; handled by specificity controls + SSIM coherence.

- **Nonlinearity** (belief encoded nonlinearly) → threatens U1 measurement validity; handled by MLP Δ and reserved DAS as extension (this is the measurement-failure mode, not a null result).
- **Oracle mismatch** (fitted dynamics wrong in Regime B) → threatens U2/U3; handled by MSD/step/turn validation and ×2/×0.5 noise sensitivity.

Experimental Setup

- **Models:** Core: MineWorld 700M_16f (frozen). Robustness: 300M_16f, 700M_32f. Fallback: Oasis-500M. No training beyond probes/small MLPs.
- **Stimuli:** MineOcclude scenes (above); magenta sheep; Regimes A and B.
- **Condition grid:** 5 occlusion durations × speed×noise decoupling grid (2×3) × 2 regimes, plus occluder-only and no-wall positive controls.
- **Baselines / floors:** Shuffled labels; random-weight MineWorld; raw-pixel probe; last-seen position heuristic; time-since-occlusion-only predictor; occluder-only scenes.
- **Ceiling:** Explicit-feature regression given [last-seen position, estimated speed, elapsed time]. (Regime A: constant velocity = oracle mean, so it is a reference; Regime B: constant velocity is a genuine baseline.)
- **Intervention controls:** Norm-matched random directions; orthogonal/perpendicular direction; wrong layer; wrong timestep; α=0; SSIM coherence.
- **Metrics:** U1: posterior/uncertainty-decoding probe R^2 vs floors. U2: partial correlation $\rho(\hat{\sigma},\sigma|\tau)$ with bootstrap CI, reliability diagram + ECE, coverage of z. U3: $\hat{\sigma}$ growth rate vs oracle, reveal-snapback paired test, absence-as-evidence tracking. U4: signed dose-response slope + bootstrap CI, permutation sign tests, specificity/coherence. $\Delta = R^2(\text{MLP}) - R^2(\text{linear})$ throughout.

Datasets and Evaluation

- **MineOcclude contents.** ~250–400 clean trajectories of 12–20 s each (~60k–120k frames) with: frames; actions; object positions (x,z) and velocities; camera pose; dual visibility labels; occlusion onset/offset; Kalman closed-form posterior mean and variance (Regime A) and particle-filter posterior (Regime B); optional posterior histograms; evaluation code. Splits are by scene layout, never splitting a trajectory across train/test. Optional scale-up to ~800–1,000 trajectories only if probes look data-starved.
- **Why a new dataset is needed.** No existing world-model benchmark ships Bayesian-oracle posteriors for occluded objects. WorldBench (2601.21282), Physics-IQ, IntPhys/InfLevel, and OccluBench score behavioral output plausibility; none provides a ground-truth posterior to grade internal beliefs against. MineOcclude is the first to make "calibrated against the Bayesian posterior" measurable.
- **Probe training data.** Per-layer mean-pooled activations from teacher-forced rollouts over MineOcclude, z-scored, with 5-fold layout-grouped CV; targets are oracle μ and log σ. Evaluation on held-out layouts.

Benchmarks/Evaluation Sets

System / work	Model type	Occlusion?	Decodes uncertainty?	Bayesian oracle?	Uncertainty-targeted steering?
Othello-GPT (Li; Nanda)	discrete seq. model	no	no	no	vector-arithmetic (state)
IRIS/DIAMOND probing (Zhang 2603.21546)	pixel RLWM	no	no	no	generic (state)
Beyond Surface Statistics (Chen)	latent diffusion	no	no	no	edit (state)
Invisible Hand of Physics (Esmati)	video diffusion	implicit	no (binary)	no	no
Shai et al. belief geometry	HMM transformer	n/a	yes (posterior)	analytic MSP	no
CAV physics steering (2605.24322)	video encoder	yes	no	no	plausibility flip
Mei et al. C3 (2512.05927)	trained DiT (not frozen)	qualitative only	trained probe (non-frozen)	no (L1 latent-error threshold)	no
This work	frozen pixel WM	yes	yes (μ+σ)	Kalman + PF	reappearance location

- **Full baseline list.**
  - **Floors:** shuffled labels, random-weight model, raw-pixel probe, last-seen position, time-since-occlusion-only, occluder-only.
  - **Reference/ceiling:** constant-velocity extrapolation (reference in A, baseline in B), explicit-feature regression.
  - **Intervention:** random/orthogonal directions, wrong layer/timestep, α=0, SSIM.

Ideal Results

Headline hypothesis: MineWorld represents calibrated posterior uncertainty about the occluded object at intermediate layers: U1–U4 hold up to roughly 1× the context window, with a graceful collapse at 1.5× (beyond context). Mechanistically, we expect (analysis a) $\hat{\mu}$ decodable from early intermediate layers with $\hat{\sigma}$ emerging slightly later in an intermediate "uncertainty emergence" band; (b) $\hat{w}_\sigma$ substantially non-collinear with the time direction $\hat{w}_\tau$ (uncertainty is evidence-conditioned, not a clock); (c) calibration degrading sharply exactly at the context edge, causally tying any miscalibration to the architectural memory limit; (d) hidden-object information localized to last-seen-position and/or occluder-region tokens and propagated frame-to-frame; and (U4) a signed dose-response confirming the estimate is used.

Alternative-results ladder:

1. **State (but not calibrated) uncertainty (U1 holds, U2 fails):** The mean is decodable but the spread is miscalibrated (overconfident). Mechanism (analysis e): distinguish frozen last-seen snapshot ($\hat{\sigma}$ flat, $\hat{\mu}$ stuck), mean-only tracking ($\hat{\mu}$ extrapolates, $\hat{\sigma}$ flat), or clock substitution ($\hat{\sigma} = f(\tau)$). Implication differs per mechanism; all are safety-relevant (overconfident exactly where blind).
2. **Represented-but-not-used (U1–U3 hold, U4 fails):** A calibrated posterior is decodable but steering it produces no signed reappearance shift. Mechanism: the representation is present at the probed layer but not read out by downstream generation — a usage/pathway failure, localizable by which layer's injection does or does not move outputs.
3. **Lack of representation / memory failure (U1 fails):** Even the mean is not decodable beyond last-seen baseline. Mechanism (analyses a, c): no emergence band appears, and collapse coincides with the context edge — the model does not maintain a hidden-object estimate; forgetting is architectural.
4. **Measurement failure (probe/oracle artifact):** Signals vanish under nonlinearity checks (MLP−linear Δ), oracle-mismatch controls, or layout-grouped CV. Mechanism: attribute to probe capacity or oracle misspecification, not model behavior; report as a negative-controls outcome.

Potential Limitations

- MineWorld was trained on VPT-style human gameplay; our controlled superflat scenes with a NoAI teleported sheep are out-of-distribution, which could depress decodability (a confound for U1), mitigated by the visible-frame positive control and the no-wall replication.
- Findings are about MineWorld (and light 300M/700M/32f checks); Genie-3 and other closed models are out of scope. We claim results for open pixel autoregressive world models, not all world models.
- Decodability ≠ use; hence U4. Linear-probe negatives are ambiguous (measurement-failure mode), addressed by MLP Δ and reserved DAS.
- Steering could induce generic damage; addressed by norm-matched/orthogonal/wrong-layer/wrong-timestep controls and SSIM coherence, plus the time-direction projection so effects can't be clock-advancing.
- Regime-A oracle is exact; Regime-B fitted dynamics could be wrong, threatening U2/U3, bounded by MSD/step/turn validation and ×2/×0.5 noise sensitivity.